

# 6

## Discussion and conclusions

---

### Intermezzo

Chapter 5 showed that students recognized LLM reasoning limitations and redistributed cognitive effort accordingly. Together, the four studies located misalignment at every information-processing stage. Chapter 6 will synthesize the findings, answer the research questions and the central question of the thesis, and derive design principles and recommendations for future research.

## 6.1 Recap: Motivation, research questions, and approach

Machines that work alongside humans have progressed from mechanical devices and teleoperated systems to AI-driven robots that perceive, reason, and act in unstructured environments shared with people (Firoozi et al., 2025). Multimodal large language models (MLLMs) and vision language action models (VLAs) now enable robots to interpret open-ended instructions, plan multi-step actions, generate motor commands from multimodal input and generate natural language explanations (Ahn et al., 2023; Black et al., 2025; Driess et al., 2023). The developments in robotics are driven by the need for higher productivity, safer workplaces, and the support for aging populations, among others (e.g., Graetz & Michaels, 2018; Gihleb et al., 2022; Acemoglu & Restrepo, 2022).

However, capable robots are not a guarantee for effective collaboration, as humans and AI-driven robots perceive, reason, and respond through different information-processing mechanisms. When human and robot processes are misaligned, interaction can fail, regardless of the robot's capability. This thesis investigated where alignment and misalignment occur between humans and AI-driven robots, and how human-robot interaction (HRI) should be designed for complementary interaction.

The thesis adopted the vocabulary of Wickens' Multiple Resource Theory (MRT). The MRT not only distinguishes processing stages (perception, cognition, responding), but also processing modalities (visual and auditory for perception), and processing codes (spatial, verbal; Wickens, 2002, 2008). In this thesis, four studies targeted different processing stages and progressed in platform complexity and realism for real-world applications.

Chapter 2 targeted the processing stage 'perception', the basis (first stage) for subsequent processing, and examined how humans and MLLMs align and differ in visual-spatial search. Chapter 3 covered all three processing stages and multiple modalities; it examined how control modality (voice vs. gesture) and mobility condition (walking vs. standing) affect performance and intuitiveness when teleoperating a mobile robot. Chapter 4 developed an open-source chess robot integrating visual perception, AI-based reasoning, and multimodal responding as a testbed. Chapter 5 focused on the cognitive stage, by investigating whether a gamified workshop can teach engineering students to recognize AI reasoning limitations and redistribute cognitive effort between themselves and the model.

The studies followed a methodological progression: controlled benchmarking with online human participants and LLMs (Chapter 2), a laboratory user study with in-person engineering students (Chapter 3), an international online survey combined with expert gameplay (Chapter 4), and cohort-scale educational deployment (Chapter 5). Participant populations ranged from 1,250 online respondents (Chapter 2) and 597 international survey participants (Chapter 4) to 218 (Chapter 3) and 226 (Chapter 5) in-person engineering students, contributing to the empirical breadth and statistical power of the studies in this thesis.

## 6.2 Main research findings

Across the four studies, a combined 2,292 human participants generated empirical evidence spanning the different stages of human-AI interaction. The following subsections summarize the methods, results, and conclusions of each chapter before synthesizing them.

### 6.2.1 Chapter 2: Human-like attention? A psychophysical comparison of visual search in humans and MLLMs

Chapter 2 tested whether MLLMs exhibit human-like attentional bottlenecks in visual-spatial search. A total of 1,250 participants, recruited via the online platform Prolific, completed nine conditions that differed in search type (feature vs. conjunction), distractor heterogeneity, and dimensionality (2D vs. 3D), yielding 157,042 valid trials across seven set sizes (4 to 256). Three MLLMs (Gemini 2.0 Flash-Lite, Gemini 2.5 Flash-Lite, and Gemma 3) were queried on the same 6,300 stimuli.

Correspondence in visual search outcomes was strong: Spearman correlations between human response times and MLLM error rates reached  $\rho = 0.91$ , and between human and MLLM error rates  $\rho = 0.82$ . Both humans and MLLMs showed efficient feature search and degraded conjunction search, consistent with Feature Integration Theory (Treisman & Gelade, 1980) and Guided Search (Wolfe, 2021).

However, the underlying mechanisms were found to differ: Humans exhibited a speed-accuracy trade-off with roughly doubled response times on target-absent trials, consistent with the quitting threshold in Guided Search 6.0 (Wolfe, 2021). The MLLMs, on the other hand, did not feature this mechanism and defaulted to biased ‘absent’ or ‘present’ responses in the more complex conditions, with high (>50%) error rates in the hardest search conditions. This deficiency in feature-binding resembles findings from recent research on feature binding as a limitation of vision-language architectures (Campbell et al., 2024; Tangtartharakul & Storrs, 2025; Mehrani & Tsotsos, 2023). The difference between human search and MLLM search can be framed as an analogy introduced earlier: Like humans do in ambient processing, MLLMs encode the entire image in a single parallel pass. Humans then search a scene through a small high-resolution fovea, using sequences of fixations and saccades that allow for strategic sampling of the image, such as first filtering on color and then serially verifying shape among remaining candidates. Most current MLLMs lack this sequential re-inspection. As a result, the overall difficulty rankings across conditions corresponded well between humans and MLLMs, but the search strategies themselves, and the error patterns they produced in complex conjunction conditions, did not.

In summary, the study of Chapter 2 contributed to literature in three ways: (1) it replicated previously established human visual search findings at scale, (2) it provided a systematic psychophysical evaluation of MLLM visual processing, and (3) it offered a direct human-MLLM comparison on identical stimuli. The main conclusion is that MLLMs learned fundamental search principles through pre-training but do not replicate the focal search strategy that humans use. This dissociation observed in Chapter 2 represents an elementary form of misalignment: two agents that appear to perceive the same scene similarly at a surface level, yet fail under

different conditions and for different reasons. This can also have practical consequences. For example, if similar perceptual dissociations extend to embodied robot vision, a warehouse robot and its human supervisor might achieve comparable object detection rates overall, yet the robot could fail specifically on targets defined by a conjunction of features. This answers Research question 1: *MLLMs do exhibit human-like attentional bottlenecks at the outcome level, but their processing diverges fundamentally from human scanning strategy, most likely due to the absence of a focal search stage and iterative feature-binding.*

### **6.2.2 Chapter 3: Walk Along: An Experiment on Controlling the Mobile Robot ‘Spot’ with Voice and Gestures**

Chapter 3 investigated how control modality (speech vs. gesture) and mobility condition (walking vs. standing) affect cognitive load, spatial reasoning, and performance. In a within-subjects 2×2 design, 218 participants teleoperated Boston Dynamics Spot along a 14-node indoor trajectory using voice vs. gesture commands while standing or walking, via the Microsoft HoloLens 2. The experiment of Chapter 3 covered all three processing stages: visual-spatial perception (monitoring the robot), spatial cognition (maintaining the robot’s pose and selecting the next command), and verbal (voice) or spatial (gesture) response.

The results showed that voice-walking was preferred by 53% of participants; gesture-standing was least favored (47% rated it last). Furthermore, voice produced faster inter-command times (5.50 s vs. 5.96 s), higher command detection ratings (4.60 vs. 4.22 on a scale of 1 “disagree” to 5 “agree”) and was preferred by 71% of participants. Walking improved intuitiveness ratings for both modalities. Gesture-standing produced significantly more left-right errors ( $M = 0.213$ ) than gesture-walking ( $M = 0.032$ ), a difference which points to so-called stimulus-response incompatibility (Fitts & Seeger, 1953). This incompatibility arises when operators mentally rotate gestures into the robot’s frame of reference (cf. Shepard & Metzler, 1971), creating a conflict when processing the spatial code used for responding and the visual-spatial code used for monitoring the robot. Voice commands avoid this conflict by operating in a verbal code, thereby reducing left-right conflict for the human. An odometry-based analysis further showed that when allowed to walk, participants physically aligned themselves with the robot more in the gesture condition (46.8%) than in the voice condition (41.9%), suggesting that the embodied nature of gesture control motivated operators to reduce spatial misalignment through repositioning.

In summary, Chapter 3 showed that misalignment is, in part, a property of the interface: command modality redistributes load across the resource pools of the operator. On the technical side, Chapter 3 contributed a validated gesture recognition pipeline for the HoloLens 2 (99% classification accuracy). Thus, the human operator’s command speed and error rate were determined by two layers: the spatial congruence between operator and robot (a human factors issue, reflected in intuitiveness ratings and left-right errors) and the accuracy with which the robot’s recognition system detected commands (a technical issue, where voice recognition was faster and more robust than gesture recognition). When allowed to walk, participants physically co-aligned with the robot to reduce spatial misalignment, particularly in the gesture condition. For robot control design, voice commands should therefore be preferred when egocentric spatial alignment cannot be upheld. Gesture control, however, should not be dismissed entirely:

a substantial minority valued it for its directness and engagement, and combining it with physical co-location largely resolved the left-right confusion observed in the stationary condition. These findings answer Research question 2: *voice control outperformed gestures on both task performance and perceived intuitiveness, and walking benefited both modalities but especially gesture control, where physical co-alignment mitigates the left-right errors observed in the stationary condition.*

### **6.2.3 Chapter 4: An Open-Source Reproducible Chess Robot for Human-Robot Interaction Research**

Chapter 4 developed OpenChessRobot, a chess robot that integrates a Franka Emika Panda robot arm, a ZED2 stereo camera, computer vision (more than 96% piece recognition accuracy), the Stockfish chess engine, and ChatGPT-generated verbal commentary. This research focused on the processing stages ‘perception’ and ‘response’: the robot perceived the chessboard visually and responded through physical manipulation and spoken verbal commentary. The game of chess was chosen because human-AI cognitive differences are well-documented in this specific task (Shannon, 1950; McCarthy, 1990; Campbell et al., 2002) and chess game outcomes are enormously diverse yet objectively verifiable.

An international survey of 597 Prolific participants evaluated the platform across three scenarios. Acceptance was highest for robotics education ( $M = 3.96$  on a scale of 1 “Definitely not” to 5 “Definitely yes”) and lowest for home entertainment ( $M = 3.59$ ), where 57% expressed negative expectations about replacing screen-based media. These findings made it clear that acceptability depends on the social role the robot fills. An expert session with a FIDE Master further showed that the LLM-based commentary lacked sufficient depth for expert-level chess. Although the LLM solved the language-fluency challenge (see e.g., Wallkötter et al., 2021) that is important for explainable embodied agents, the content of the commentary lacked the strategic understanding of the game, i.e., it projected confidence that exceeded the model’s actual understanding of the game. This illustrates a form of response misalignment: verbal fluency masks a lack of domain reasoning. Together, these results answer Research question 3: *the chess robot serves as a testbed for human-AI collaboration research, and users valued AI-generated verbal interaction in educational contexts but found it strategically shallow at the expert level.*

### **6.2.4 Chapter 5: Educating future engineers about LLMs: A scalable workshop**

Chapter 5 focused on the processing stage ‘cognition’: the human’s ability to recognize AI reasoning limitations and compensate accordingly. The premise was that, as LLMs become more capable, the human needs to do less of the reasoning, whereas with less capable models, the human must provide more guidance, hints, or even solve parts of the task directly. A key learning objective for the future is therefore for engineering students to recognize how much of the cognitive work they need to do themselves and how much they can delegate to an LLM, a skill that will only grow in importance as LLMs continue to evolve.

In ten workshop sessions, 226 MSc engineering students prompted a non-reasoning LLM (Gemini 2.0 Flash-Lite) and a reasoning LLM (Gemini 2.5 Pro) to solve grid navigation tasks of

increasing complexity, with plans executed in simulation and on a Boston Dynamics Spot robot. The study engaged the cognition stage with both spatial and verbal codes: students formulated verbal prompts encoding spatial goals and evaluated spatial plans.

The results showed that the reasoning model achieved 56% penalty-free solutions on easy tasks versus 19% for the non-reasoning model, but declined to 22% on the hardest task, consistent with findings that LLM reasoning quality degrades with complexity (Shojaee et al., 2025). Students changed their prompting accordingly: 49% used step-by-step instructions for the non-reasoning model, while on the hardest task with the reasoning model only 3% did so; 37% instead provided waypoints and 23% strategic heuristics.

Post-workshop data ( $n = 200$ ) showed that 91.0% of participants reported high engagement and 81.5% reported substantial learning, consistent with research on ‘active learning’ in the engineering education domain (Bonwell & Eison, 1991; Freeman et al., 2014; Prince, 2004). Our gamification approach was used to facilitate learning through game-based incentives (Morales-Trujillo & García-Mireles, 2020; Sailer & Homner, 2020). Chapter 5 demonstrated that misalignment can represent a learning opportunity: experiencing LLM boundaries in a scorable task taught students to redistribute cognitive load, reflecting the ‘complementary’ aim of the overarching research question. This answers Research question 4: *through such a scalable, gamified workshop, students could recognize LLM reasoning limitations and redistribute cognitive effort accordingly, as evidenced by their adaptive prompting strategies and high self-reported learning.*

### **6.2.5 Synthesis and misalignment as a unifying construct**

The four chapters studied the alignment problem from perception to integrative cognition, and revealed that misalignment can take place in different forms, at each information-processing stage. Chapter 2 revealed perceptual misalignment: outcome-level alignment coexisted with divergent visual encoding mechanisms. Chapter 3 revealed spatial cognition-response misalignment: stimulus-response incompatibility where gesture control interfered with visual monitoring more than voice did. Chapter 4 revealed verbal response misalignment: surface-level verbal coherence masked shallow reasoning. Chapter 5 revealed cognitive misalignment: human and AI reasoning processes differed on the same planning tasks: humans relied on spatial intuition such as heuristics and waypoints, while the LLM relied on token-level sequential generation; and students initially failed to compensate for this difference in their prompts.

A number of overarching observations can be made across these forms. First, *outcome-level metrics alone are insufficient for detecting misalignment*: high correlations in Chapter 2 masked different search strategies, fluent chess commentary in Chapter 4 concealed shallow reasoning, and penalty-free rates in Chapter 5 obscured the fact that the reasoning model’s performance degraded sharply with complexity while failure modes changed from collisions to planning breakdowns.

Second, *as the robot/AI system becomes more automated, the human role changes from operator to supervisor*: from directly commanding every movement (Chapter 3), through

evaluating automated perception and strategy (Chapter 4), to diagnosing and scaffolding complete LLM-generated plans (Chapter 5). This progression points to the need for educational interventions that prepare users for the supervisory role by making robot/AI failure modes visible. In addition, since humans and MLLMs can differ in how they perceive the same scene (Chapter 2), predicting where the AI fails can be difficult for human supervisors. This thesis illustrates how the transition from manual control to supervisory control, a concept historically studied extensively in tightly controlled domains such as control rooms and aviation (e.g., Carvalho et al., 2008; Sarter & Woods, 1995), now reveals itself in modern LLM-driven human-robot interaction.

Third, *the quality of human-robot collaboration depends on how cognition is distributed, not on the absolute capability of either partner*. Voice control in Chapter 3 freed spatial resources while gestures overloaded them; Chapter 5 extended this principle to the reasoning level, where students redistributed cognitive work based on model type and task difficulty. Chapter 2 revealed the risk when distribution is misaligned at the perceptual stage: if a supervisor assumes the AI perceives as they do, they may fail to verify the conditions under which the model errs.

These observations complement classical frameworks of human-machine interaction. The Fitts list (Fitts, 1951), Moravec's paradox (Moravec, 1988), and Sheridan and Verplank's (1978) levels-of-automation framework all assumed static allocation of capabilities between human and machine. Later refinements allowed adjustment along levels and stages of automation (Parasuraman et al., 2000). *The present thesis shows that static allocation is inadequate for LLM-driven robots*: LLM capabilities are context-dependent and partially unpredictable (Chapters 2 and 5), and the human mental model of AI capability is dynamic and often initially miscalibrated (Chapter 5). Function allocation therefore has to be dynamic and learned in real time. Unlike classical adaptive automation, in which the system adjusts its own automation level in response to operator state (Parasuraman, 2001; Kaber & Endsley, 2004), the present findings call for operator-led task allocation (also called adaptable automation; Sauer et al., 2012): the human actively redistributes cognitive work based on an evolving model of AI competence. In Chapter 5, students fulfilled this role, redistributing cognitive work according to observed model performance and task complexity.

The above observations culminate in an answer to the central question posed in Section 1.1: *where do the processes of humans and AI-driven robots align, where do they diverge, and how can interaction be designed so that those differences become complementary?* Human and AI-driven robot processes **align at the level of outcomes yet diverge at the level of underlying mechanisms**, and this divergence can emerge at every information-processing stage as well as at the interface that couples the two agents. The differences **become complementary when function allocation is dynamic and operator-led**: the human redistributes cognitive work based on an evolving model of AI competence as task demands and observed AI performance change.

A further consideration is that a transparency asymmetry exists: humans can introspect on their limitations and signal cognitive overload, whereas current LLMs can hardly indicate when they operate beyond their competence (Kim et al., 2025; also see Chapter 4 and 5), making misalignment detection largely dependent on the human side. Future research may want to distinguish between misalignment that improved AI capability can resolve (such as binding deficits [Chapter 2] or shallow chess commentary [Chapter 4]) and structural misalignment, such as spatial reference-frame conflicts (Chapter 3) or the inherent opacity of LLM reasoning.

### 6.3 Practical contributions

Apart from the scientific findings described above, this thesis produced practical outputs for reuse by researchers and practitioners. Each chapter delivered open tools, datasets, or educational materials. These include a psychophysical benchmark and large-scale human dataset (Chapter 2), an open-source mixed-reality control pipeline (Chapter 3), a reproducible chess robot with open hardware designs (Chapter 4), and a scalable workshop package with curriculum materials (Chapter 5). The four chapters enable other groups to replicate or build upon this work.

Chapter 2 produced a benchmark of 6,300 synthetic visual search images across nine conditions grounded in visual search theory (Treisman & Gelade, 1980; Wolfe, 2021), extending the CLEVR framework (Johnson et al., 2017). The human dataset of 157,042 trials from 1,250 participants is one of the largest psychophysical visual search datasets to date. The benchmark enables standardized comparison of any MLLM against human perceptual performance, with relevance to tasks such as robotic inspection and assistive vision applications.

Chapter 3 contributed an open-source pipeline for voice and gesture control of the Boston Dynamics Spot via the HoloLens 2, including a custom SVM-based gesture recognition module and ROS-based communication architecture. The study also yielded design guidelines for multimodal robot control, such as preferring voice when egocentric spatial alignment is unavailable and supporting physical co-location for gesture interfaces. Methodologically, Chapter 3 introduced LLM-based content analysis of participant interview transcripts using ChatGPT o1 pro mode, validated against manual coding ( $r = 0.98$ ). This approach was subsequently adopted in Chapters 4 and 5 for analyzing open-ended survey responses and student prompts, respectively, forming a methodological thread across the thesis. Code and data are publicly available.

Chapter 4 produced the *OpenChessRobot*, the first open-source chess robot designed for reproducible HRI research. Source code, 3D-printed gripper designs, trained models, and data are publicly available. Reproducibility remains a challenge in HRI (Gunes et al., 2022), and open hardware enables cumulative progress in robotics (Patel et al., 2023). Without shared platforms, researchers have to rebuild experimental setups from scratch, which makes it difficult to compare findings across studies or build on prior work. Open-source systems enable other research teams to replicate experiments under comparable conditions. The modular design allows component substitution without redesigning the system.

Chapter 5 produced a reusable workshop package: a web-based platform, simulation environment, gamification framework, instructor guides, and a three-task curriculum, all openly available. Delivery to 226 students at \$23 total API cost demonstrated scalability. As LLMs become standard engineering tools, curricula that teach their capabilities and limitations, not only their use, will be essential (Kasneci et al., 2023; Lo, 2023).

In addition to producing functional prototypes, the four studies collectively yielded a number of viable design principles. First, designers should use modality switching to offload overloaded cognitive resource pools: Chapter 3 showed that shifting from gesture to voice freed the operator's spatial resources, reducing left-right errors. Second, interfaces should expose AI confidence/uncertainty rather than only outcomes: Chapters 2 and 4 showed that outcome-level metrics and verbal fluency can mask underlying failures, so displaying mechanism-level information (e.g., feature-binding difficulty, reasoning depth) would help operators anticipate where the AI is likely to fail. Third, calibration training should occur before deploying the robot: Chapter 5 demonstrated that even a short, gamified intervention shifted students from over-reliance to adaptive task allocation, suggesting that structured exposure to AI failure modes can build realistic expectations. Fourth, function allocation should be designed as dynamic and operator-led rather than fixed a priori: because LLM capabilities are context-dependent and partially unpredictable (Chapters 2 and 5), the human needs both the authority and the information to redistribute cognitive work in real time as task demands and AI performance change.

## 6.4 Limitations

This thesis has several limitations. Firstly, this study focused heavily on the visual processing modality; several other processing modalities are unexplored in this thesis. In particular, the haptic modality was not investigated, despite its growing importance in tasks such as teleoperation (Popken et al., 2023; Pacchierotti & Prattichizzo, 2024), dexterous manipulation (Kappasov et al., 2015; Zhao et al., 2025), assistive care (Nanavati et al., 2024), collaborative manipulation (Mol et al., 2026), and humanoid robotics in general (Sanctuary AI, 2025). The auditory modality was used in this thesis (see Chapter 3) but not systematically studied. For example, this thesis did not examine how prosody, timing, or turn-taking in robot speech affects cognitive alignment, though these factors are known to affect conversational grounding (Mavridis, 2015; Skantze, 2021).

The thesis also assumed modularized and verifiable robot cognitive stages. End-to-end perception-to-execution architectures such as vision-language-action models (Black et al., 2025; Brohan et al., 2023) fuse these stages into a single pipeline with non-interpretable intermediate representations, challenging the stage-based categorization adopted in this thesis. Rapid model improvements may reduce the bottlenecks documented here (feature binding, spatial planning), but structural differences between human and AI cognition are unlikely to disappear.

Another limitation of this thesis is that each study used a specific domain and participant population. Chapter 2 used synthetic arrays with geometric shapes (squares and cubes), which is rather far from the rich scenes robots encounter in practice. Chapter 3 studied one robot (Spot), one device (HoloLens 2), and a young and technically educated sample (mean age 23.5 years). Chapter 4 focused on chess, a complete-information domain, unlike typical HRI where information is partial. Chapter 5 again recruited engineering students at the same university, a population likely above average in AI literacy.

The progression from controlled online benchmarking (Chapter 2) through a laboratory experiment (Chapter 3) and an international survey combined with expert gameplay (Chapter 4) to ecological deployment in a university course (Chapter 5) gradually sacrificed internal validity for increased ecological validity. More specifically, Chapter 2 allowed for precise parameter manipulation but used artificial stimuli. Chapter 3 introduced embodied interaction but involved a relatively simple trajectory for the robot. Chapter 4 gained breadth through an international survey but did not feature a longitudinal design. Chapter 5 achieved full realism within an actual university course but could not assess long-term retention of skills or knowledge. The four studies therefore differ in measures, populations, and control, making the synthesis in Section 6.2.5 primarily qualitative.

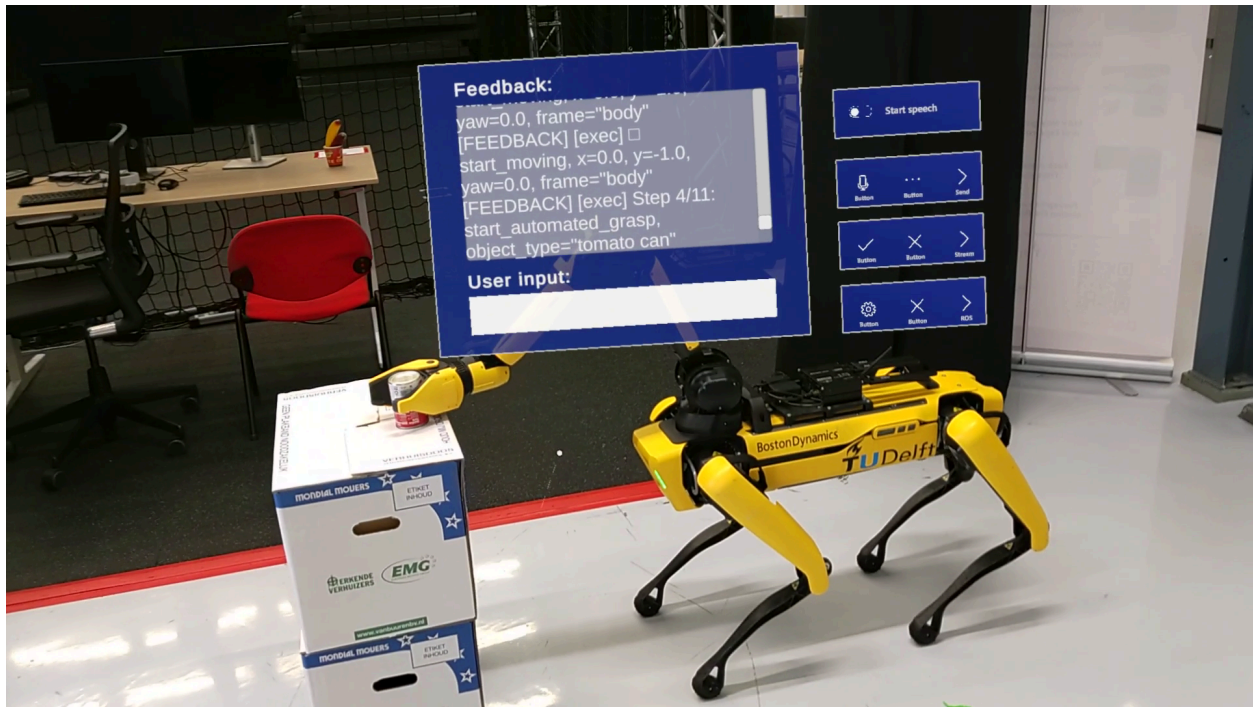
In summary, whether misalignment transfers to more complex, unstructured environments (e.g., retail, outdoor, home), other HRI domains (e.g., healthcare, manufacturing, domestic robotics) and populations (older adults, non-experts, non-engineers) requires further investigation. Therefore, it should be acknowledged that the claim of misalignment occurring at any MRT stage is domain-general in principle but validated only within these domains.

## **6.5 Recommendations for future research**

The limitations above suggest several directions for future research. Systematic studies of haptic and auditory modalities could examine whether tactile feedback reduces the visual-spatial overload in Chapter 3 (Culbertson et al., 2018; Wang et al., 2024) and how prosody and turn-taking in robot speech affect conversational grounding (Skantze, 2021).

The detection of pieces by the chess robot (see Chapter 4) relied on standard computer vision techniques. In a follow-up study by an MSc student whom the author of this dissertation co-supervised (Waterhout, 2024), more advanced pose estimation methods for the chess pieces were used, together with synthetic training data (3D chess pieces rendered via Blender, a similar approach to that used for generating the 3D shapes in Chapter 2). As AI has improved, MLLMs have become capable of recognizing complex and ill-defined objects, such as “a cube in a cluttered environment” (Chapter 2) or a “can of tomatoes” (as demonstrated in Chapter 5). Another MSc student (Verbaan, 2024), co-supervised by the author of this dissertation, developed a system that used an MLLM to interpret camera images of a tabletop scene, identify manipulable objects, and decompose natural language user instructions into executable action sequences for the Franka Emika robotic arm. Chapter 5 took this further, by combining MLLM-driven perception with navigation in a single pipeline. Future work should extend this line

of research to more demanding real-world settings, with systematic investigation of long-tail error patterns.



**Figure 6.1.** Prototype head-mounted mixed-reality interface combining verbal commands, LLM-generated planning feedback, and holographic overlays of robot execution. The user uttered the command “Pick up the tomato can” and is monitoring the robot’s execution progress.

Apart from improvements on the robot side, future work should also augment the human operator. Chapter 3 used a HoloLens mixed reality device to provide visual overlays of issued commands. Additionally, in a prototype of the setup in Chapter 5 (see Figure 6.1), we extended the HoloLens setup so that it combines verbal input, LLM verbal feedback, and holographic display of robot feedback, creating an interaction mode comparable to the simulation interface in Chapter 5 but with real-time visual overlays. Augmented reality also allows for testing virtual robots or setups in an immersive manner before deploying the same robots or setups in the real world. For example, the author co-conducted a study that used a mixed-reality device to test different ways of displaying machine-state information on large virtual screens (Tabone et al., 2026). As wearable augmented reality devices become more accessible, such augmentation could help operators calibrate their mental model of robot capabilities by making robot perception and planning visible (Wang et al., 2026), to better address the misalignment documented across this thesis.

Furthermore, the neurosymbolic approach of Chapter 4, coupling a symbolic problem solver with an LLM for explanation, is a promising direction for explainable AI (e.g., an AI-based chess tutor that can explain why it made a particular move). High-quality chess reasoning requires move accuracy, which modern chess engines achieve through decades of developments in tree-search-based methods such as Stockfish. However, engine evaluations do not directly

translate to explanation quality from a learner's perspective, and collecting expert human feedback is substantially harder than the preference labelling used for general-purpose LLMs. Advances in LLM capabilities may partially address this (cf. Kolasani et al., 2025), but domain-expert evaluation data or pipelines are still needed to calibrate LLMs that genuinely explain strategic reasoning rather than produce superficially fluent but strategically shallow verbal commentary.

The stage-based framework as used in the Introduction of this thesis could be extended or complemented with elements suited to human-AI resource sharing. Shared mental model theory for human-autonomy teams (Demir et al., 2020) and interdependence analysis (Johnson et al., 2014) may offer such complementary elements. Another possible extension is modeling how humans update their mental models of AI capability over extended interaction, drawing on Bayesian trust models (Guo & Yang, 2021) or computational theory of mind (Rabinowitz et al., 2018).

As foundation models improve in feature binding and spatial reasoning, research should track whether misalignment is actually resolved, or just shifted to subsequent processing stages. Longitudinal studies comparing the same tasks across model generations would be needed to provide empirical data to test the robustness of our findings. Additionally, it is recommended that the misalignment concept be tested in applied HRI domains other than education (such as healthcare and manufacturing), and with more diverse populations. Finally, the practical contributions point toward valorisation opportunities: the gamified workshop could be adapted for professional training where misalignment stakes are higher, and the open-source platforms facilitate such translational work.

## **6.6 Broader reflections and outlook**

Across the studies, the human's cognitive role changes: from direct teleoperation (Chapter 3), through a robot serving as opponent, teacher, and partner (Chapter 4, where participants valued it most in educational roles), to supervision (goal-setting and verification) (Chapter 5). As AI capability improves in the future, this transition will likely accelerate toward even more supervisory control and less direct control. In other words, as AI assumes more perception and cognition, the human role moves to meta-cognitive monitoring. Chapter 5 showed this empirically: students offloaded spatial-verbal planning to the AI and reallocated resources to verification. Whether this enhances system performance depends on humans maintaining situation awareness to detect AI failures.

The above implications extend to education and the labor market. Future engineers need training not only in AI tools but also in maintaining cognitive engagement during AI-assisted problem-solving, detecting reasoning failures, and knowing when to override. Productivity gains from AI-driven robots will not be distributed equally: workers with automatable tasks may face displacement (Frey & Osborne, 2017; Acemoglu & Restrepo, 2020), while those who can supervise and calibrate AI are more likely to benefit. The dividing line may be the ability to recognize and manage misalignment, a competence not yet systematically taught in current

education. It can be expected that in the future, detecting errors, hallucinations, or even sabotage by highly advanced AI will be a challenging but critically important task for humanity.

The studies in this thesis revealed several risks when AI takes on cognitive roles. MLLMs produced confident but incorrect visual search output (Chapter 2), the reasoning model generated hallucinated navigation plans and therefore would require an extra safety layer for avoiding collisions in real-life applications (Chapter 5; Ravichandran et al., 2026), and the commentary of the chess robot projected confidence exceeding its understanding (Chapter 4). Current LLMs do not reliably signal when they operate beyond their competence, placing the cost of opacity on the human. Trust calibration (Lee & See, 2004; Parasuraman & Riley, 1997; Guo & Yang, 2021) becomes increasingly important and difficult when AI competence boundaries are not transparent. Students initially overestimated LLM capabilities (Chapter 5), and in safety-critical domains, users may not have the opportunity to learn AI competence boundaries through training or repeated use (Tomsett et al., 2020).

Based on this evidence, it is argued that meaningful human control is necessary. If AI development disregards human authority and understanding of the machine, the misalignment documented in Chapters 2 through 5 would be seen as irrelevant rather than a problem to solve. That framing might yield short-term efficiency but would remove the feedback loop through which humans detect errors and maintain accountability. Human oversight should therefore be designed into systems through transparent interfaces that display uncertainty, training that builds calibrated trust (Chapter 5), and institutional accountability. The EU AI Act (European Parliament, 2024) and trustworthy AI standards (Floridi et al., 2018; Jobin et al., 2019) provide a regulatory structure, but translating these into interaction designs is still an open challenge. The trajectory from direct control (Chapter 3) to goal-setting (Chapter 5) showed that human work will change from performing tasks to supervising them. This need not mean unemployment, but the benefits of human-AI collaboration may go primarily to those trained to manage it, while others become passive recipients of AI decisions. Education (Chapter 5), policy, professional standards, and technology design have to work together.

The empirical research in this thesis represents an increment in understanding human-AI collaboration, but the implications extend beyond the domains studied in this thesis. Misalignment occurs even when aggregate AI performance matches humans (Chapter 2), which indicates that benchmark-only evaluation can be misleading. Interface design affects misalignment independently of agent capability (Chapter 3), which suggests that regulatory attention should include interaction design. Short educational interventions can improve AI calibration (Chapter 5), a finding that supports investment in AI literacy.

This thesis was itself shaped by the rapid evolution it studies. At the start of this PhD in mid-2022, large generative models had not yet entered mainstream use. The release of ChatGPT five months later prompted the LLM-based commentary in the chess robot (Chapter 4, the earliest study) and raised the multimodal interaction questions behind Chapters 2 and 3. The emergence of reasoning-capable LLMs in 2024–2025 inspired a vision-to-plan pipeline, but early experiments revealed perception as the bottleneck, redirecting the work toward Chapter 2

and informing Chapter 5. LLMs also transformed how the author conducted this research, from writing support to co-coding, making the judgment of when to rely on AI and when to override it (also see Chapter 5) a daily practice rather than an abstract research topic.

## 6.7 Conclusions

This thesis investigated where human and AI-driven robot cognition aligns, where it misaligns, and how interaction can make differences complementary. Four empirical studies with a total of 2,292 participants, spanning the perceptual, cognitive, and response stages of information-processing, revealed that misalignment does occur and is consequential. Agents that appear to perceive the same scene fail under different task conditions, the human-robot interface determines where cognitive conflicts arise, surface coherence in AI outputs can conceal shallow reasoning, and humans initially miscalibrate their expectations of AI capability. The overarching insight is that function allocation between human and AI in complex environments has to be continuously recalibrated through interaction.

On the design side, this thesis recommends a number of principles: use modality switching to offload overloaded processing codes, expose AI confidence and uncertainty rather than only outcomes, provide calibration training before deployment of the robot, and design human vs. AI function allocation as dynamic and operator-led. On the research side, the recommendations are to extend the investigation to unexplored processing modalities (haptic, auditory), test human-robot misalignment in other applied domains, and scale educational approaches that build the cognitive competence for human-AI collaboration.

As AI-driven robots assume increasingly diverse roles, the human contribution will change from direct control to supervision. Whether this transition strengthens or weakens human agency depends on investment in transparent interaction design, calibrated trust, and training and access to skills for effective human-AI collaboration. The path from ‘misalignment’ to ‘collaboration’ requires deliberate design, empirical validation, and ongoing effort.

## References

- Acemoglu, D., & Restrepo, P. (2020). Robots and jobs: Evidence from US labor markets. *Journal of Political Economy*, 128(6), 2188–2244. <https://doi.org/10.1086/705716>
- Acemoglu, D., & Restrepo, P. (2022). Demographics and automation. *The Review of Economic Studies*, 89(1), 1–44. <https://doi.org/10.1093/restud/rdab031>
- Ahn, M., Brohan, A., Brown, N., Chebotar, Y., Cortes, O., David, B., Finn, C., Fu, C., Gopalakrishnan, K., Hausman, K., Herzog, A., Ho, D., Hsu, J., Ibarz, J., Ichter, B., Irpan, A., Jang, E., Jauregui Ruano, R., Jeffrey, K., ... Zeng, A. (2023). Do as I can, not as I say: Grounding language in robotic affordances. In K. Liu, D. Kulic, & J. Ichnowski (Eds.), *Proceedings of the 6th Conference on Robot Learning* (Vol 205, pp. 287–318). PMLR. <https://proceedings.mlr.press/v205/ichter23a/ichter23a.pdf>
- Black, K., Brown, N., Driess, D., Esmail, A., Equi, M., Finn, C., Fusai, N., Groom, L., Hausman, K., Ichter, B., Jakubczak, S., Jones, T., Ke, L., Levine, S., Li-Bell, A., Mothukuri, M., Nair, S., Pertsch, K., Shi, L., ... Zhilinsky, U. (2025).  $\pi_0$ : A vision-language-action flow model for general robot control. *Robotics: Science and Systems XXI*, Los Angeles, CA. <https://doi.org/10.15607/rss.2025.xxi.010>
- Bonwell, C. C., & Eison, J. A. (1991). *Active learning: Creating excitement in the classroom* (ASHE-ERIC Higher Education Report No. 1). ERIC Clearinghouse on Higher Education. <https://files.eric.ed.gov/fulltext/ED336049.pdf>

- Brohan, A., Brown, N., Carbajal, J., Chebotar, Y., Chen, X., Choromanski, K., Ding, T., Driess, D., Dubey, A., Finn, C., Florence, P., Fu, C., Arenas, M. G., Gopalakrishnan, K., Han, K., Hausman, K., Herzog, A., Hsu, J., Ichter, B., ... Zeng, A. (2023). RT-2: Vision-language-action models transfer web knowledge to robotic control. In J. Tan, M. Toussaint, & K. Darvish (Eds.), *Proceedings of the 7th Conference on Robot Learning* (Vol 229, pp. 2165–2183). PMLR. <https://proceedings.mlr.press/v229/zitkovich23a/zitkovich23a.pdf>
- Campbell, M., Hoane, A. J., Jr., & Hsu, F. (2002). Deep Blue. *Artificial Intelligence*, 134(1–2), 57–83. [https://doi.org/10.1016/S0004-3702\(01\)00129-1](https://doi.org/10.1016/S0004-3702(01)00129-1)
- Campbell, D., Rane, S., Giallanza, T., De Sabbata, N., Ghods, K., Joshi, A., Ku, A., Frankland, S. M., Griffiths, T. L., Cohen, J. D., & Webb, T. (2024). Understanding the limits of vision language models through the lens of the binding problem. *Advances in Neural Information Processing Systems* 37 (pp. 113436–113460). NeurIPS. <https://doi.org/10.52202/079017-3604>
- Carvalho, P. V., dos Santos, I. L., Gomes, J. O., Borges, M. R., & Guerlain, S. (2008). Human factors approach for evaluation and redesign of human–system interfaces of a nuclear power plant simulator. *Displays*, 29(3), 273–284. <https://doi.org/10.1016/j.displa.2007.08.010>
- Culbertson, H., Schorr, S. B., & Okamura, A. M. (2018). Haptics: The present and future of artificial touch sensation. *Annual Review of Control, Robotics, and Autonomous Systems*, 1, 385–409. <https://doi.org/10.1146/annurev-control-060117-105043>
- Demir, M., McNeese, N. J., & Cooke, N. J. (2020). Understanding human-robot teams in light of all-human teams: Aspects of team interaction and shared cognition. *International Journal of Human-Computer Studies*, 140, Article 102436. <https://doi.org/10.1016/j.ijhcs.2020.102436>
- Driess, D., Xia, F., Sajjadi, M. S. M., Lynch, C., Chowdhery, A., Ichter, B., Wahid, A., Tompson, J., Vuong, Q., Yu, T., Huang, W., Chebotar, Y., Sermanet, P., Duckworth, D., Levine, S., Vanhoucke, V., Hausman, K., Toussaint, M., Greff, K., Zeng, A., Mordatch, I., & Florence, P. (2023). PaLM-E: An embodied multimodal language model. *Proceedings of the 40th International Conference on Machine Learning*, 202 (pp. 8469–8488). PMLR. <https://proceedings.mlr.press/v202/driess23a/driess23a.pdf>
- European Parliament. (2024). Regulation (EU) 2024/1689 of the European Parliament and of the Council laying down harmonised rules on artificial intelligence (Artificial Intelligence Act). *Official Journal of the European Union*, L 2024/1689. <https://eur-lex.europa.eu/eli/reg/2024/1689/oj/en>
- Firooz, R., Tucker, J., Tian, S., Majumdar, A., Sun, J., Liu, W., ... Schwager, M. (2025). Foundation models in robotics: Applications, challenges, and the future. *The International Journal of Robotics Research*, 44(5), 701–739. <https://doi.org/10.1177/02783649241281508>
- Fitts, P. M. (Ed.). (1951). *Human engineering for an effective air-navigation and traffic-control system*. National Research Council, Division of Anthropology and Psychology. <https://apps.dtic.mil/sti/tr/pdf/ADB815893.pdf>
- Fitts, P. M., & Seeger, C. M. (1953). S-R compatibility: Spatial characteristics of stimulus and response codes. *Journal of Experimental Psychology*, 46(3), 199–210. <https://doi.org/10.1037/h0062827>
- Floridi, L., Cowls, J., Beltrametti, M., Chatila, R., Chazerand, P., Dignum, V., Luetge, C., Madelin, R., Pagallo, U., Rossi, F., Schafer, B., Valcke, P., & Vayena, E. (2018). AI4People: An ethical framework for a good AI society. *Minds and Machines*, 28, 689–707. <https://doi.org/10.1007/s11023-018-9482-5>
- Freeman, S., Eddy, S. L., McDonough, M., Smith, M. K., Okoroafor, N., Jordt, H., & Wenderoth, M. P. (2014). Active learning increases student performance in science, engineering, and mathematics. *Proceedings of the National Academy of Sciences*, 111(23), 8410–8415. <https://doi.org/10.1073/pnas.1319030111>
- Frey, C. B., & Osborne, M. A. (2017). The future of employment: How susceptible are jobs to computerisation? *Technological Forecasting and Social Change*, 114, 254–280. <https://doi.org/10.1016/j.techfore.2016.08.019>
- Gihleb, R., Giuntella, O., Stella, L., & Wang, T. (2022). Industrial robots, workers' safety, and health. *Labour Economics*, 78, Article 102205. <https://doi.org/10.1016/j.labeco.2022.102205>
- Graetz, G., & Michaels, G. (2018). Robots at work. *The Review of Economics and Statistics*, 100(5), 753–768. [https://doi.org/10.1162/rest\\_a\\_00754](https://doi.org/10.1162/rest_a_00754)
- Gunes, H., Broz, F., Crawford, C. S., Rosenthal-von der Pütten, A., Strait, M., & Riek, L. (2022). Reproducibility in human-robot interaction: Furthering the science of HRI. *Current Robotics Reports*, 3, 281–292. <https://doi.org/10.1007/s43154-022-00094-5>

- Guo, Y., & Yang, X. J. (2021). Modeling and predicting trust dynamics in human-robot teaming: A Bayesian inference approach. *International Journal of Social Robotics*, 13, 1899–1909. <https://doi.org/10.1007/s12369-020-00703-3>
- Jobin, A., Ienca, M., & Vayena, E. (2019). The global landscape of AI ethics guidelines. *Nature Machine Intelligence*, 1, 389–399. <https://doi.org/10.1038/s42256-019-0088-2>
- Johnson, J., Hariharan, B., van der Maaten, L., Fei-Fei, L., Zitnick, C. L., & Girshick, R. (2017). CLEVR: A diagnostic dataset for compositional language and elementary visual reasoning. Proceedings of the 2017 IEEE Conference on Computer Vision and Pattern Recognition, Honolulu, HI, 1988–1997. <https://doi.org/10.1109/CVPR.2017.215>
- Johnson, M., Bradshaw, J. M., Feltovich, P. J., Jonker, C. M., van Riemsdijk, M. B., & Sierhuis, M. (2014). Coactive design: Designing support for interdependence in joint activity. *Journal of Human-Robot Interaction*, 3(1), 43–69. <https://doi.org/10.5898/JHRI.3.1.Johnson>
- Kaber, D. B., & Endsley, M. R. (2004). The effects of level of automation and adaptive automation on human performance, situation awareness and workload in a dynamic control task. *Theoretical Issues in Ergonomics Science*, 5(2), 113–153. <https://doi.org/10.1080/1463922021000054335>
- Kappassov, Z., Corrales, J. A., & Perdureau, V. (2015). Tactile sensing in dexterous robot hands. *Robotics and Autonomous Systems*, 74, 195–220. <https://doi.org/10.1016/j.robot.2015.07.015>
- Kasneci, E., Sessler, K., Küchemann, S., Bannert, M., Dementieva, D., Fischer, F., Gasser, U., Groh, G., Günemann, S., Hüllermeier, E., Krusche, S., Kutyniok, G., Michaeli, T., Nerdel, C., Pfeffer, J., Poquet, O., Sailer, M., Schmidt, A., Seidel, T., Stadler, M., Weller, J., Kuhn, J., & Kasneci, G. (2023). ChatGPT for good? On opportunities and challenges of large language models for education. *Learning and Individual Differences*, 103, Article 102274. <https://doi.org/10.1016/j.lindif.2023.102274>
- Kim, S. S. Y., Wortman Vaughan, J., Liao, Q. V., Lombrozo, T., & Russakovsky, O. (2025). Fostering appropriate reliance on large language models: The role of explanations, sources, and inconsistencies. *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, Article 420. ACM. <https://doi.org/10.1145/3706598.3714020>
- Kolasani, S., Saplin, M., Crispino, N., Montgomery, K., Davis, J. Q., Zaharia, M., Wang, C., & Wang, C. (2025). LLM CHESS: Benchmarking reasoning and instruction-following in LLMs through chess [Paper presentation]. First Workshop on the Foundations of Reasoning in Language Models (FoRLM) at NeurIPS 2025, San Diego, CA, United States. <https://openreview.net/forum?id=i0UHRCKEXZ>
- Lee, J. D., & See, K. A. (2004). Trust in automation: Designing for appropriate reliance. *Human Factors*, 46(1), 50–80. [https://journals.sagepub.com/doi/10.1518/hfes.46.1.50\\_30392](https://journals.sagepub.com/doi/10.1518/hfes.46.1.50_30392)
- Lo, C. K. (2023). What is the impact of ChatGPT on education? A rapid review of the literature. *Education Sciences*, 13(4), Article 410. <https://doi.org/10.3390/educsci13040410>
- Mavridis, N. (2015). A review of verbal and non-verbal human-robot interactive communication. *Robotics and Autonomous Systems*, 63, 22–35. <https://doi.org/10.1016/j.robot.2014.09.031>
- McCarthy, J. (1990). Chess as the Drosophila of AI. In T. A. Marsland & J. Schaeffer (Eds.), *Computers, chess, and cognition* (pp. 227–237). Springer. [https://doi.org/10.1007/978-1-4613-9080-0\\_14](https://doi.org/10.1007/978-1-4613-9080-0_14)
- Mehrani, P., & Tsotsos, J. K. (2023). Self-attention in vision transformers performs perceptual grouping, not attention. *Frontiers in Computer Science*, 5, Article 1178450. <https://doi.org/10.3389/fcomp.2023.1178450>
- Mol, N., Prendergast, J. M., Abbink, D. A., & Peternel, L. (2026). Fitts' list revisited: An empirical study on function allocation in a two-agent physical human-robot collaborative position/force task. *IEEE Robotics and Automation Letters*, 11(1), 202–209. <https://doi.org/10.1109/LRA.2025.3632607>
- Morales-Trujillo, M. E., & García-Mireles, G. A. (2020). Gamification and SQL: An empirical study on student performance in a database course. *ACM Transactions on Computing Education*, 21(1), Article 3. <https://doi.org/10.1145/3427597>
- Moravec, H. (1988). *Mind children*. Harvard University Press. <https://dl.acm.org/doi/abs/10.5555/48030>
- Nanavati, A., Ranganeni, V., & Cakmak, M. (2024). Physically assistive robots: A systematic review of mobile and manipulator robots that physically assist people with disabilities. *Annual Review of Control, Robotics, and Autonomous Systems*, 7, 123–147. <https://doi.org/10.1146/annurev-control-062823-024352>
- Pacchierotti, C., & Prattichizzo, D. (2024). Cutaneous/tactile haptic feedback in robotic teleoperation: Motivation, survey, and perspectives. *IEEE Transactions on Robotics*, 40, 978–998. <https://doi.org/10.1109/TRO.2023.3344027>

- Parasuraman, R. (2001). Adaptive automation: From theory to practice. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, 45(4), 291–292. <https://doi.org/10.1177/154193120104500405>
- Parasuraman, R., & Riley, V. (1997). Humans and automation: Use, misuse, disuse, abuse. *Human Factors*, 39(2), 230–253. <https://doi.org/10.1518/001872097778543886>
- Parasuraman, R., Sheridan, T. B., & Wickens, C. D. (2000). A model for types and levels of human interaction with automation. *IEEE Transactions on Systems, Man, and Cybernetics-Part A: Systems and Humans*, 30(3), 286–297. <https://doi.org/10.1109/3468.844354>
- Patel, V. V., Liarakapis, M. V., & Dollar, A. M. (2023). Open robot hardware: Progress, benefits, challenges, and best practices. *IEEE Robotics & Automation Magazine*, 30(3), 123–148. <https://doi.org/10.1109/MRA.2022.3225725>
- Popken, M., Prendergast, J. M., Wiertelowski, M., & Peternel, L. (2023). An adaptive semi-autonomous impedance controller for teleoperated object grasping based on human grip safety margin. *2023 IEEE-RAS 22nd International Conference on Humanoid Robots (Humanoids 2023)* (pp. 1–8). IEEE. <https://doi.org/10.1109/Humanoids57100.2023.10375144>
- Prince, M. (2004). Does active learning work? A review of the research. *Journal of Engineering Education*, 93(3), 223–231. <https://doi.org/10.1002/j.2168-9830.2004.tb00809.x>
- Rabinowitz, N. C., Perbet, F., Song, F., Zhang, C., Eslami, S. M. A., & Botvinick, M. (2018). Machine theory of mind. In J. Dy & A. Krause (Eds.), *Proceedings of the 35th International Conference on Machine Learning* (Vol 80, pp. 4218–4227). PMLR. <https://proceedings.mlr.press/v80/rabinowitz18a.html>
- Ravichandran, Z., Robey, A., Kumar, V., Pappas, G. J., & Hassani, H. (2026). Safety guardrails for LLM-enabled robots. *IEEE Robotics and Automation Letters*, 11(4), 4649–4656. <https://doi.org/10.1109/LRA.2026.3667488>
- Sailer, M., & Homner, L. (2020). The gamification of learning: A meta-analysis. *Educational Psychology Review*, 32(1), 77–112. <https://doi.org/10.1007/s10648-019-09498-w>
- Sanctuary AI. (2025). *Sanctuary AI equips general purpose robots with new touch sensors for performing highly dexterous tasks*. <https://www.sanctuary.ai/blog/sanctuary-ai-equips-general-purpose-robots>
- Sarter, N. B., & Woods, D. D. (1995). How in the world did we ever get into that mode? Mode error and awareness in supervisory control. *Human Factors*, 37(1), 5–19. <https://doi.org/10.1518/001872095779049516>
- Sauer, J., Kao, C. S., & Wastell, D. (2012). A comparison of adaptive and adaptable automation under different levels of environmental stress. *Ergonomics*, 55(8), 840–853. <https://doi.org/10.1080/00140139.2012.676673>
- Shannon, C. E. (1950). XXII. Programming a computer for playing chess. *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science*, 41(314), 256–275. <https://doi.org/10.1080/14786445008521796>
- Shepard, R. N., & Metzler, J. (1971). Mental rotation of three-dimensional objects. *Science*, 171(3972), 701–703. <https://doi.org/10.1126/science.171.3972.701>
- Sheridan, T. B., & Verplank, W. L. (1978). *Human and computer control of undersea teleoperators*. Massachusetts Institute of Technology, Man-Machine Systems Lab. <https://doi.org/10.21236/ADA057655>
- Shojaee, P., Mirzadeh, I., Alizadeh, K., Horton, M., Bengio, S., & Farajtabar, M. (2025). The illusion of thinking: Understanding the strengths and limitations of reasoning models via the lens of problem complexity. *Advances in Neural Information Processing Systems* 38. <https://openreview.net/forum?id=YghiOusmvw>
- Skantze, G. (2021). Turn-taking in conversational systems and human-robot interaction: A review. *Computer Speech & Language*, 67, Article 101178. <https://doi.org/10.1016/j.csl.2020.101178>
- Tabone, W., De Winter, J., Zhang, R., Wickens, C. D., van de Ruit, M., & Eisma, Y. B. (2026). *Embodied sampling: An augmented reality investigation of visual attention* [Preprint]. ResearchGate. [https://www.researchgate.net/profile/Joost-De-Winter/publication/395535610\\_Embodied\\_sampling\\_An\\_augmented\\_reality\\_investigation\\_of\\_visual\\_attention/links/6970c5bc52773b62239c5082/Embodied-sampling-An-augmented-reality-investigation-of-visual-attention.pdf](https://www.researchgate.net/profile/Joost-De-Winter/publication/395535610_Embodied_sampling_An_augmented_reality_investigation_of_visual_attention/links/6970c5bc52773b62239c5082/Embodied-sampling-An-augmented-reality-investigation-of-visual-attention.pdf)
- Tangtharakul, G., & Storrs, K. R. (2025). Visual language models show widespread visual deficits on neuropsychological tests. *Nature Machine Intelligence*, 8, 209–219. <https://doi.org/10.1038/s42256-026-01179-y>

- Tomsett, R., Preece, A., Braines, D., Cerutti, F., Chakraborty, S., Srivastava, M., Pearson, G., & Kaplan, L. (2020). Rapid trust calibration through interpretable and uncertainty-aware AI. *Patterns*, 1(4), Article 100049. <https://doi.org/10.1016/j.patter.2020.100049>
- Treisman, A. M., & Gelade, G. (1980). A feature-integration theory of attention. *Cognitive Psychology*, 12(1), 97–136. [https://doi.org/10.1016/0010-0285\(80\)90005-5](https://doi.org/10.1016/0010-0285(80)90005-5)
- Verbaan, L. E. (2024). *Perception and control with large language models in robotic manipulation: Developing and assessing an integrated large language model system on environmental and task complexity* [Master's thesis, Delft University of Technology]. TU Delft Repository. <https://resolver.tudelft.nl/uuid:2944a8e1-0a54-492b-af63-193a7ac11db8>
- Wallkötter, S., Tulli, S., Castellano, G., Paiva, A., & Chetouani, M. (2021). Explainable embodied agents through social cues: A review. *ACM Transactions on Human-Robot Interaction*, 10(3), Article 27. <https://doi.org/10.1145/3457188>
- Wang, G., Wang, H. H., & Ren, G. (2024). Visual and haptic guidance for enhancing target search performance in dual-task settings. *Applied Sciences*, 14(11), Article 4650. <https://doi.org/10.3390/app14114650>
- Wang, H., Phatak, R., Ocampo, J., & Han, Z. (2026). Indicating robot vision capabilities with augmented reality. *International Journal of Social Robotics*, 18(3), Article 42. <https://doi.org/10.1007/s12369-026-01368-0>
- Waterhout, M. (2024). *Evaluating 6D pose estimation accuracy with synthetic data: A comparative analysis of RGB-based and RGB-depth-based images in a chess piece picking task with a robotic arm* [Master's thesis, Delft University of Technology]. TU Delft Repository. <https://resolver.tudelft.nl/uuid:4506c969-220d-4257-ad5b-91853ea83ccb>
- Wickens, C. D. (2002). Multiple resources and performance prediction. *Theoretical Issues in Ergonomics Science*, 3(2), 159–177. <https://doi.org/10.1080/14639220210123806>
- Wickens, C. D. (2008). Multiple resources and mental workload. *Human Factors*, 50(3), 449–455. <https://doi.org/10.1518/001872008X288394>
- Wolfe, J. M. (2021). Guided Search 6.0: An updated model of visual search. *Psychonomic Bulletin & Review*, 28(4), 1060–1092. <https://doi.org/10.3758/s13423-020-01859-9>
- Zhao, Z., Li, W., Li, Y., Liu, T., Li, B., Wang, M., ... Zhu, S. C. (2025). Embedding high-resolution touch across robotic hands enables adaptive human-like grasping. *Nature Machine Intelligence*, 7(6), 889–900. <https://doi.org/10.1038/s42256-025-01053-3>